Madhav Kumar Yadav

Aspiring Data Scientist | ML Engineer (Python, MLOps)

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Professional Summary

Final-year B.Tech (CSE) student focused on applied machine learning and MLOps. Built end-to-end ML projects in Python covering regression, classification and forecasting, from EDA and model comparison to Flask/FastAPI services, Docker, MLflow tracking and cloud deployment. Earned the **WorldQuant University Data Science Modeling** credential (Credly) and built a Streamlit public-health analytics platform. Currently learning Deep Learning and advanced MLOps; seeking a Data Science / ML Engineer internship.

Education

Jagannath University | *B.Tech, Computer Science & Engineering* 2023 – 2027

CGPA: 7.7/10 • Coursework: Machine Learning, Data Structures & Algorithms, DBMS, Probability & Statistics, Big Data Analytics

Projects

HealthSentinel – Public Health Analytics Platform | *Streamlit, MongoDB, Python* GitHub | Live Demo

- Built an 8-page Streamlit platform (Springboard virtual internship) over 25K+ records in a star-schema model, with an ETL cleaning pipeline, optional MongoDB Atlas storage, and PDF/JSON report export.
- Added ARIMA outbreak forecasting, IQR anomaly flags, KMeans risk clustering, a report-grounded Q&A assistant and salted-hash login; containerized with Docker.

Network Security – Phishing Detection Pipeline | *Scikit-learn, MLflow, FastAPI* GitHub

- Built a modular pipeline (MongoDB ingestion, schema validation, KNN imputation, training) on 11K+ records with 30 features; tuned 5 classifiers with grid search and selected a 256-tree Random Forest.
- Reached ≈96.7% test accuracy (F1 ≈0.97); logged runs to MLflow + DagsHub; served batch predictions via FastAPI; Dockerized with a GitHub Actions CI workflow.

Student Performance Predictor | *Scikit-learn, CatBoost, Flask* GitHub

- Built an installable ML pipeline (ingestion, transformation, training, prediction) with custom logging/exceptions; compared 9 regressors (incl. CatBoost, XGBoost) on 1,000 records.
- Best model reached test R^2 ≈0.88 (RMSE ≈5.4 points); deployed through a Flask app with Docker and AWS Elastic Beanstalk config.

Forest Fire Weather Index (FWI) Prediction | *Flask, Scikit-learn* GitHub | Live Demo

- Performed EDA and correlation-based feature selection on the Algerian Forest Fires data; compared Linear, Lasso, Ridge and ElasticNet with 5-fold CV (LassoCV, RidgeCV, ElasticNetCV).
- Deployed a Ridge model with StandardScaler (test R^2 ≈0.98, MAE ≈0.69) as a Flask web app on Render.

MLOps Practice: Airflow & Experiment Tracking | *Airflow, MLflow, Docker* GitHub

- Orchestrated a data pipeline with Apache Airflow in Docker; practiced MLflow autologging and DagsHub experiment tracking.

Technical Skills

Languages	Python, SQL
ML / Data	Scikit-learn, CatBoost, XGBoost, Pandas, NumPy, Statsmodels (ARIMA), Matplotlib, Seaborn, EDA
MLOps / Backend	Docker, MLflow, DagsHub, Apache Airflow, Flask, FastAPI, GitHub Actions (CI/CD)
Databases	MySQL, MongoDB, SQLite
Visualization / Tools	Power BI, Tableau, Streamlit, Excel, Git, GitHub, Jupyter, VS Code
Deployment	Render, AWS Elastic Beanstalk, Streamlit Cloud

Certifications & Training

- **WorldQuant University – Data Science Modeling** (Credly badge): four hands-on projects across CSV, SQLite and MongoDB covering data cleaning and EDA, correlation analysis, Ridge/Lasso regression, classification evaluation (accuracy, precision, recall, confusion matrix) and AR/ARMA time-series forecasting.
- **Springboard Virtual Internship** – Healthcare Operations Intelligence (analytics and dashboarding)
- **Master Microsoft Power BI: Beginner to Data Analyst Pro** – Udemy
- **Python Essentials 1** – Cisco Networking Academy (Jul 2025); **Introduction to Cybersecurity** – Cisco (Aug 2024)